

A Correlated Signals

Computational Game Theory

Model of Litigation Bargaining

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Abstract

A linear programming algorithm solves extensive-form game trees representing litigation games in which parties receive correlated quality signals. Litigants make file, answer, offer, and quit decisions. The model allows examination of interactions among different game features and parameters. With baseline parameters, including risk neutrality, the parties are strikingly aggressive in their settlement demands, and settlement is rare. The effects of fee shifting depend markedly on whether fee shifting is triggered only at trial or also by nonanswer or pretrial withdrawal.

1 Introduction

A natural assumption for settlement bargaining models is that the plaintiff and defendant receive correlated signals of the quality of the litigation. In a tort case, for example, each party may possess a combination of shared and private information about the events leading to an accident, and so each party can be understood as receiving a noisy signal of the true merits of the case. Even in a case in which facts are unambiguous and the only dispute is as to the law, each party will assess the case based on its lawyer's analysis and hunches, and the parties' estimates of case strength based on such legal advice will be correlated but not necessarily identical. Most of the literature modeling settlement bargaining, however, does not feature correlated signals. In the classic models of Landes (1971) [16], Posner (1973) [20], and Gould (1973) [11], parties' estimates of litigation quality are entirely exogenous. Later Bayesian models, such as Bebchuk (1984) [3] and Polinsky

and Rubinfeld (1998) [19], assume that one party knows the quality of litigation while the other party knows the distribution of all cases but does not receive a signal of the particular case's quality.

Instead of correlated information, some models grant the plaintiff some information and the defendant some other information. The aggregate of these pieces of information then determines case quality. For example, Daughety and Reinganum (1994) [8] assign one party information on liability and another party information on damages. Similarly, in Friedman and Wittman (2006) [9], Dari-Mattiacci and Saraceno (2020) [6], and Abramowicz (2025) [1], the plaintiff and defendant each receive independent random signals from a known distribution, and the true value of the case depends on an average of those signals. Thus, in these models, each of the plaintiff's and defendant's signals correlates with the merits, but unrealistically, the signals do not correlate with one another. The game thus becomes like a game of poker in which there are no common cards.

Plausibly, the optimal strategy and resulting equilibria in such a game might be quite different from those in a game of correlated signals. An intuition is that correlated signals would lead to high rates of settlement and relatively limited opportunities for strategic negotiation behavior. With independent signals, each litigating party must worry greatly about adverse selection; if the other party agrees to settle, that may well be because the other party has received a signal that the first party's claim is actually quite strong. But with correlated signals, there is only a small probability that the other party has a highly divergent view of the case strength. A reasonable strategy then might be to make demands or offers that rise roughly linearly with case strength, effectively agreeing roughly to share the surplus from settlement. Settlement might still fail because of tail estimates, but cases that do fail should resolve at roughly their expected value.

Only one pair of articles allows the plaintiff and defendant to estimate the quality of the litigation based on correlated signals. Lee and Klerman (2016) [17] and Klerman, Lee, and Liu (2018) [15] use correlated signals to update the canonical selection model of Priest and Klein (1984) [21]. In these updated models, the true value of the litigation is Y (distributed uniformly over the entire real line), and the plaintiff and defendant receive signals of $Y + \epsilon_p$ and $Y + \epsilon_d$, respectively, where each ϵ is a normally distributed random variable. The papers differ in that the older adopts the Landes-Posner-Gould assumption that the cases settle whenever there exists a settlement amount at which each party prefers to settle, while the more recent explicitly models the bargaining structure. The later paper's bargaining model is that of Chatterjee and Samuelson (1983) [4], in which each party separately makes an offer and settlement occurs at the midpoint if the plaintiff's offer is less than or equal to the defendant's.

These papers represent significant advances in the realism of litigation models. But they have limitations that make it difficult to assess how correlated signals would impact issues other than the selection of cases for litigation. First, even in the more recent paper, the litigation quality is constrained to a range where

each party has a credible threat to take the case to trial (pp. 392-93). In effect, the plaintiff always files, the defendant always answers, and neither party ever quits before trial. Yet we have known since Shavell (1982) [22] and P'ng (1983) [18] about the importance of filing and answer decisions and since Huang (2004) [12] about the strategic importance of the plaintiff's option to abandon a suit. Second, the entire cost of litigation is thus borne all at once, rather than sequentially. Yet expenditures on litigation are dispensed over time. Third, although case quality varies, there is a simple cutoff, above which the plaintiff has a high fixed probability of winning and below which the probability is fixed at a low value. More realistically, the probability of winning may be close to 50% in some cases and a range of high or low values in others.

In this article, I relax these assumptions through use of a computational model that identifies Nash equilibria in large extensive form game trees, applying the linear programming algorithm identified in von Stengel, van den Elzen, and Talman (2002) [24]. Abramowicz (2025) applies this algorithm to model settlement bargaining with game trees of up to 16,111 nodes but does not consider correlated signals of litigation quality. Rather, that article seeks to establish a proof of concept for computational (or algorithmic) game theory modeling by using a setup similar to that of Dari-Mattiacci and Saraceno, in which the parties have independent signals that aggregate to determine case quality. The ten-offer specifications in the present article have 45,211 nodes, while the fifteen-offer extension has 97,211 nodes.

Despite this greater complexity, the solutions identified in this article remain perfect Bayesian Nash equilibria. That is, the plaintiff and defendant each assign a mixed strategy to each information set, and each player's mixed strategy is a best response for each player given that player's information. That is, neither player could improve its strategy even by a small ϵ by changing its strategy, holding the other player's strategy constant, in the modeled game. Perfection guarantees that a player's choices remain optimal if a decision is actually reached, even at decision points that are not reached in equilibrium, given beliefs satisfying the equilibrium's consistency requirements. The mixed strategies are exact rational numbers, though payoffs are rounded to the nearest 10^{-5} of the potential judgment to speed calculation.

The algorithmic game theory model thus follows in the Bayesian tradition of recent settlement bargaining scholarship, but with enhanced strategic complexity, including multiple levels of case quality, multiple signal levels, variable approaches to fee-shifting, adjustable levels of risk aversion, and litigant options to file, answer, and quit before trial. Though a computational model may not offer the elegance and generalizability of analytical models, the richness of game features can provide a useful complement to those models. The computational approach shines in three ways. First, it can illustrate how different game features may interact when analytical computation might be intractable. Second, the computational approach can make it possible to explore relatively small changes in applicable rules without developing an entirely new model. Third, welfare effects may be precisely measured, and different aspects of accuracy may be distinguished.

The comparison of fee-shifting rules illustrates all of these advantages. Switching from our baseline American rule simulation to a regime in which fees are shifted after trial makes the plaintiff more conciliatory at one signal consistent with a slightly weak case, producing an increase in settlement but leaving the rest of the equilibrium largely intact. Extending fee shifting to cases in which the defendant does not answer or a party withdraws, however, has a much more substantial effect. The direct effect of the change is that a defendant with a weak case is paradoxically more likely to answer, because a defendant who does not answer must not only pay damages, but also the plaintiff's filing fees. That in turn discourages plaintiff filing. Expansion of fee shifting thus adversely affects meritorious plaintiffs while benefiting truly not liable defendants.

This is not a just so story, but a set of conclusions that can be supported by considering equilibria and how each party's best response would change when parts of the opponent's strategy are taken (say, filing behavior) from one equilibrium and parts, from another. For example, we can assess how much of an observed change in a party's behavior with a change in a fee-rule would occur if the other party retained the strategy that it had absent the change, as well as if the other party changed to its new equilibrium strategy in some ways but not others. The model not only reveals an intuition, that extending fee shifting to nonanswer makes defendants more willing to answer, but also can show that the magnitude of this effect may outweigh competing effects. This analysis does not conclusively resolve the question of what stages should be included under a fee-shifting rule, let alone the broader debate concerning the advisability of the American and British rules, but it helps identify policy stakes that might otherwise be quite challenging to model, given the interdependence of entrance and exit decisions.

These results relate to a literature showing how the timing of litigation expenditures shapes bargaining by changing the credibility of threats. Bebchuk (1996) [2] shows how dividing costs across stages of litigation can make what would be a negative-expected-value claim if fully litigated viable, while Croson and Mnookin (1996) [5] and Hubbard (2016) [13] show how litigants can enhance the credibility of the threat to continue pursuing a claim or defense by sinking expenditures in advance. Central to this logic is the option to exit before trial. The greater the extent to which a party has already incurred costs before the exit decision and thus reduced the costs remaining to be incurred, the less attractive exit becomes. That in turn makes the party's continued litigation more credible in earlier settlement bargaining.

The wrinkle added by extending fee shifting to exit is that if exiting itself would force a party to pay the opponent's costs incurred to date, that also ultimately makes the party's continued litigation more credible. This can counteract the strategic advantage that the plaintiff may otherwise gain from incurring costs before the defendant. When the plaintiff spends money to file and a fee-shifting rule requires the defendant to reimburse the cost if it does not answer, then answering preserves the possibility of avoiding that liability,

and further, its own expenses in answering might end up being borne by the plaintiff. The option of exiting before trial, meanwhile, becomes less attractive to both parties, since unilateral exit requires reimbursement of the other party's already-incurred costs. While it is not entirely intuitive how much these effects will matter, the results indicate that they are of vital significance. While plaintiffs often do not file, defendants always answer, in both the risk neutral setting and with risk aversion, including in robustness checks assessing the possibility of multiple equilibria. It is this dynamic that ultimately redounds on average to defendants' benefit when fee shifting applies in cases of unilateral exit.

The computational results also offer many intriguing observations along the way. The parties are highly aggressive bargainers with risk neutrality, not so much deciding how much of the perceived settlement surplus they wish to insist on, but instead holding out for outcomes far more favorable than their assessments of case strength would suggest. Offers and demands often remain constant across multiple signal levels before jumping at a threshold. Substantial bargaining failure occurs even though parties have relatively good information and even though the one-round structure eliminates opportunities to influence an opponent's offer through a bluff. Meanwhile, equilibria with greater settlement do not necessarily offer lower expenditures or greater accuracy. Finally, risk aversion can increase settlement while also increasing expenditures and causing settlement values to depart further from the amounts warranted by underlying liability, particularly in the absence of fee shifting.

2 The Model

This part frames the correlated signals model in explicitly game theoretic terms by describing the extensive-form game tree. Section 2.1 describes the signals the parties and the court receive of the case quality. Section 2.2 sets out the sequence of litigation decisions and explains how the parties' information changes as the game progresses. Section 2.3 specifies how litigation costs are accrued and assigned under the three fee-shifting rules, as well as how risk preferences affect utility. Section 2.4 summarizes model parameters and illustrates a path through the game tree. Section 2.5 defines social welfare outcome measures.

2.1 Information structure

The underlying strength of the plaintiff's case, Q , is defined on a continuous range, distributed uniformly over $[0, 1]$ in the baseline simulations. Because the computational approach used here depends on extensive-form game trees, however, the game presented to the solver must have finitely many information sets and actions.

Let $\mathcal{S}_P$, $\mathcal{S}_D$, and $\mathcal{S}_C$ be the signal sets for the plaintiff, defendant, and court, and let $\mathcal{O}$ be each party's offer set. The random signals X_P , X_D , and X_C take values in the corresponding sets. The baseline has $|\mathcal{S}_P| = |\mathcal{S}_D| = 10$, $|\mathcal{S}_C| = 2$, and $|\mathcal{O}| = 10$. (As a robustness check, an extension uses $|\mathcal{O}| = 15$ for a core subset of parameters while retaining the same signal sets.)

A model with discrete signals and offers is admittedly less general in these respects than a model based on continuous quantities, and a virtue of the mathematical literature on settlement bargaining is that some variables can be modeled continuously. In Bebchuk (1984)[3], for example, the defendant knows the probability that the court will find liability, which can be any value from a mutually known probability density function. But not all variables in mathematical settlement models are continuous. In their review of the settlement bargaining literature, Daughety and Reinganum (2012) [7] include both two-types and continuous types models. Even models that offer continuous types in one respect feature only one or two types in another respect. For example, Klerman, Lee, and Liu (2018) [15] allow each party to obtain a continuous signal of the quality of litigation, but the objective probability of prevailing at trial can take on only one of two values. Thus, the conventions of the existing literature do not mandate continuous variables. The ten signal levels for each party allow a considerably greater degree of granularity than two-types models, while underlying case quality varies continuously, as in Klerman, Lee, and Liu.

The discretized signals of case quality that the plaintiff, defendant, and court receive are independent conditional on the exact value of Q . They are correlated signals, however, because each depends on Q . A party that receives a signal indicating that the plaintiff has a relatively strong case should conclude that the

other party and the court will be more likely to also have such a signal or a nearby one. The level of noise is parameterized. Specifically, the parameters σ_P , σ_D , and σ_C represent the standard deviations of the normal noise used to generate the plaintiff's, defendant's, and court's signals. That is, for a given underlying case quality $Q = q$, each noisy assessment is drawn from a normal distribution centered on q , with the standard deviation specified for the party or court.

The interval $[0, 1]$ is divided into equal-width signal intervals, and the ten party-signal labels are the interval midpoints, from 0.05 to 0.95. Noisy assessments that lie outside $[0, 1]$ are excluded. Thus, with ten signal intervals, an assessment between 0.3 and 0.4 produces the signal labeled 0.35. Conditional on any q , the probability that a party receives a particular signal is the area under the normal density curve within that signal's interval divided by the total area between zero and one.

Because the underlying case quality is continuous, calculating the probabilities used in the game for discrete signals requires integrating over the possible values of Q . For $i \in \mathcal{S}_P$ and $j \in \mathcal{S}_D$, let $p_i(q) = \Pr(X_P = i \mid Q = q)$ and $d_j(q) = \Pr(X_D = j \mid Q = q)$ denote the corresponding conditional signal probabilities. The signals are independent conditional on q , so, under the baseline uniform distribution of quality,

$$\Pr(X_P = i, X_D = j) = \int_0^1 p_i(q) d_j(q) dq.$$

This integral accounts for every possible underlying quality that could produce the observed pair of signals. This makes it straightforward to calculate the probability of any defendant signal given a particular plaintiff signal. The integrals over case quality are evaluated numerically using 64-point Gauss–Legendre quadrature, which approximates each integral by a weighted sum of evaluations at selected quality values. For the baseline information structure, an automated test verifies that increasing the number of quadrature points from 64 to 128 changes no calculated signal or true-liability probability by more than 10^{-9} .

Figure 1 illustrates the relationship between case quality and the parties' information. Because Q is defined continuously, the groupings on the left are illustrative only. The highlighted fan out demonstrates the relative probabilities that a plaintiff or a defendant will receive different signals given that the continuous case quality is somewhere between 0.6 and 0.8. The informal intent is for information to be fairly strong. The combined probability that the signal is within this range (either 0.65 or 0.75) is 39.9%, and the probability that the signal is less than 0.5 is only 17.9%.

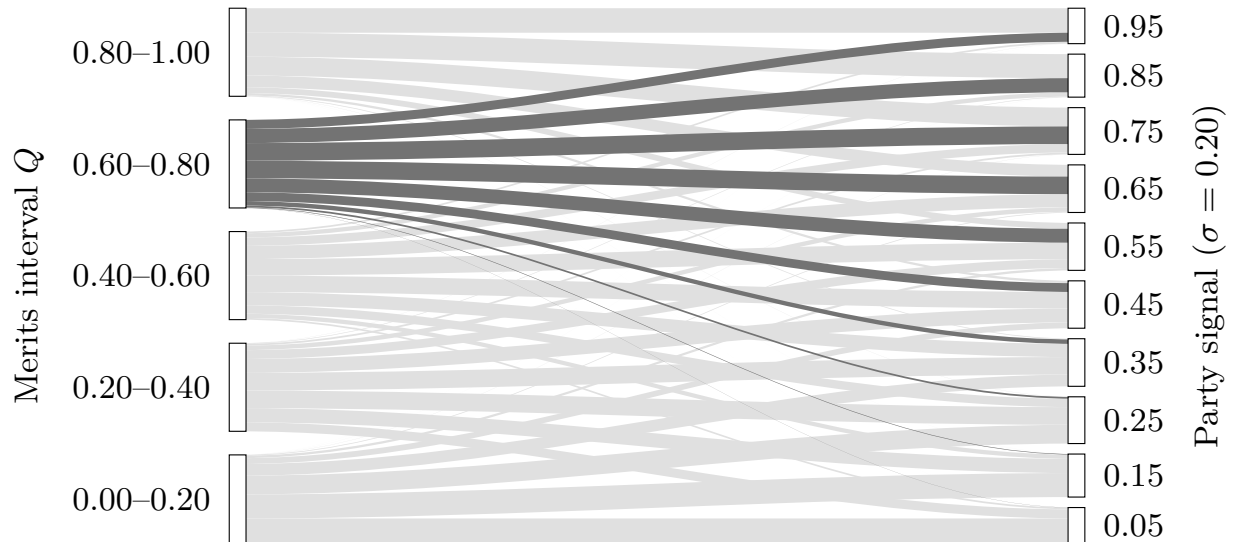


Figure 1: Information structure

A simpler design would restrict q to $\{0, 1\}$ and then to derive each party's and the court's signal directly from the true state. The approach of allowing a continuous Q is more consistent with the literature, particularly Klerman, Lee, and Liu, though they allow factual strength to vary across the entire real line while constraining trial-win probability to only two values. More significantly, continuous merits better exhibit an important feature of correlated signals, that a party expects that its opponent likely has a signal close to the party's own. Not so with a direct signal of truth. For example, when a party receives a signal of exactly 0.5 (as would be possible with an odd number of signals), the opponent is no more likely to receive a signal of 0.5 than if the party had drawn any other signal. The approach of drawing a signal directly from the underlying truth is, however, used as a robustness check and reported in Online Appendix [SPECIFY].

2.2 Litigation sequence and updating

The game begins with the plaintiff's decision whether to file and, conditional on filing, the defendant's decision whether to answer. Neither party knows its opponent's signal. Although the game tree imposes an ordering, making it appear that the Chance decision determining the plaintiff's signal occurs before the Chance decision determining the defendant's signal, that ordering has no strategic effect. An information set groups decision points that a player cannot distinguish. When the plaintiff makes its decision whether to file, it arrives at an information set containing solely its signal. This may follow any of $|\mathcal{S}_D|$ defendant game nodes. The defendant observes filing and can infer information from it. At later stages, the parties condition on both filing and answering. Thus, their strategies reflect both their private signals and the information

conveyed by participation, although no new evidentiary signal arrives during the game.

Assuming that the plaintiff files and the defendant answers, then the plaintiff faces a decision whether the plaintiff will abandon the litigation rather than incur trial costs if settlement fails, and the defendant similarly faces a decision whether to default in the event settlement fails. Neither party's decision has any immediate effect, and neither party finds out about the other party's decision. The placement of these decisions before settlement rather than after is admittedly counterintuitive. But it greatly speeds up the algorithm for calculating equilibria, relative to a model in which each player takes into account both its own settlement offer and its opponent's in deciding whether to quit. With the latter approach, there would be one information set for each player for each triple of a liability signal, a settlement offer, and an opponent's settlement offer. By deciding abandonment or defaulting in advance, the number of information sets in the game is greatly reduced, with just one information set concerning quitting for each liability signal for a player. This specification does not allow a party to revise its exit decision in response to the opponent's observed offer. Including decisions whether a party will abandon or quit litigation continues to serve its primary role of affecting bargaining indirectly, by forcing each party to consider in making a settlement offer whether its opponent has a credible threat to take a case to trial.

The bargaining protocol used is that introduced by Chatterjee and Samuelson (1983) [4] and used in Friedman and Wittman [9], Dari-Mattiacci and Saraceno (2020) [6], and Abramowicz (2025) [1]. Each party submits an offer without observing the other's, and the case settles if and only if the defendant's offer is at least as high as the plaintiff's demand, with the settlement value equal to the midpoint. While future work might employ some other more complex bargaining protocol, such as back-and-forth offers between the parties, that would increase the number of information sets and the computational burden. Chatterjee-Samuelson bargaining avoids the simplification in the Landes-Gould-Posner bargaining models of assuming that a case will settle whenever that is in the mutual interest of the parties, given their expectations. With Chatterjee-Samuelson bargaining, parties have some incentive to shade their offers in their own favor in the hope of capturing more of the settlement surplus, though they must balance this objective with the risk of bargaining failure. The approach thus allows for an important dimension of strategic bargaining, while avoiding the simplification adopted by models that arbitrarily choose a player to make a take-it-or-leave-it offer to the other player.

If a case does not settle but one party previously decided to quit in the event of settlement failure, the game ends without trial. If both parties decided to quit, a lottery assigns each a 50% chance of being the party that quits. This resolves mutual withdrawal once, rather than treating both parties as separately losing. If neither party quits after settlement failure, each incurs the additional trial cost and the court determines liability.

If bargaining fails and neither party quits, then a trial occurs. The court has two signal intervals, based on whether assessments lie below or above 0.5 and corresponding to no liability versus liability. Like the parties, the court can therefore make mistakes relative to true liability. The court’s decision rule does not refine its inference based on the parties’ filing, answering, or bargaining decisions. Making the court an additional strategic player that chooses a decision rule to optimize social welfare, more in the spirit of the mechanism-design literature such as Spier (1994) [23], remains a possible extension.

2.3 Costs, preferences, and fee shifting rules

A plaintiff prevailing at trial receives a unit damages award. The allocation of costs depends on the fee shifting rule, and the parties’ utilities depend on specification of risk preferences.

The ordinary-cost specifications set $c_{file} = c_{answer} = c_{trial} = 0.15$. Here c_{trial} is the additional trial cost for each party, so each spends 0.30 if the case reaches trial. There is no separate bargaining cost. Costs are fixed rather than chosen strategically. The model does not differentiate whether these costs take the form of attorney fees, court costs, or litigants’ time. Each party’s initial wealth is 10.0; with risk neutrality, this initial level does not affect the equilibrium.

A computational model also can easily consider the effects of risk aversion. Utility with constant absolute risk aversion is modeled as a function of wealth, $U = -e^{-\alpha W}$. The main risk-averse comparison sets $\alpha = 2$ for both parties. Utility values may be positively scaled and shifted without affecting the equilibrium. The Online Repository’s Supplemental Materials folder contains a subfolder “Risk aversion utility curves” with a diagram illustrating utility as a function of wealth with this level of risk aversion.

Under the American rule, each party bears its own costs. Trial Fee-Shifting requires the trial loser to reimburse the winner’s incurred costs. Complete Fee-Shifting adds reimbursement of the plaintiff’s filing cost when the defendant does not answer, and reimbursement of the opponent’s incurred costs when a party later exits unilaterally. No fee award attaches to nonfiling, and withdrawal does not incur future trial expenses. In settled cases, each party bears its own costs.

The justification for studying both trial fee-shifting and complete fee-shifting rules is not that different jurisdictions adopt different approaches. To the contrary, the English rule as practiced in the United Kingdom is a complete fee-shifting rule. See, e.g., *Field v Del Vecchio* [2022] EWHC 1117 (Ch), [32]–[34] [CHECK CITE]. Even in the United States, when a statute allows fee shifting to the prevailing party, a party that prevails by default qualifies. See, e.g., *Kirtsaeng v. John Wiley & Sons, Inc.*, 579 U.S. 197, 208 n.3 (2016). Trial fee-shifting, however, has sometimes been assumed in the literature as the applicable rule. Farmer & Pecorino (YEAR) [CITE – Costly Voluntary Disclosure] and Huang (2004) assume that trial expenses

only are shifted. It is worth assessing whether that assumption matters and whether there might be any advantages to fee-shifting rules that apply only after trial.

Table 1 summarizes the model primitives and fee rules.

2.4 Parameter Values and Example Path

Table 1 summarizes the parameters of the model.

Table 1: Model primitives

Primitive	Core specifications
Case quality	Unobserved continuous quality $Q \sim U[0, 1]$
True liability	$T = 1$ means truly liable. Given quality q , $\Pr(T = 1 \mid Q = q) = q$; the unconditional probability is $\Pr(T = 1) = 0.5$.
Signal generation	Each party and the court receive a noisy assessment drawn from a normal distribution centered on q , truncated to $[0, 1]$. Assessments are independent conditional on exact q .
Signal noise	$\sigma_P = \sigma_D = \sigma_C = 0.20$
Party signals	$X_P \in \mathcal{S}_P$ and $X_D \in \mathcal{S}_D$, with $ \mathcal{S}_P = \mathcal{S}_D = 10$. Ten equal-width intervals per party have midpoint labels $0.05, 0.15, \dots, 0.95$.
Court signal and rule	$X_C \in \mathcal{S}_C$, with $ \mathcal{S}_C = 2$. The court finds no liability for assessments below 0.5, and liability for assessments at or above 0.5.
Damages	Fixed award $d = 1$ if P wins at trial or D defaults; zero if P abandons or loses at trial
Initial wealth	$w_P = w_D = 10$
Filing and answering	P chooses whether to file; after filing, D chooses whether to answer or default. Nonfiling ends the dispute without payment or litigation costs.
Bargaining rounds	Single simultaneous round
Offer grid	$\mathcal{O} = \{0.05, 0.15, \dots, 0.95\}$, with $ \mathcal{O} = 10$ choices per party. The finer-grid check uses $ \mathcal{O} = 15$ midpoint offers with the same signal sets.
Settlement rule	Settlement occurs if P's demand is no greater than D's offer; D pays P the midpoint of the two amounts.
Exit	Before offers, each party privately commits to continue or quit if bargaining fails. P may abandon; D may default. Trial occurs only if bargaining fails and neither quits.
Mutual exit	If both commit to quit and bargaining fails, a lottery selects P's abandonment or D's default with probability 0.5 each.
Stage costs	$c_{file} = c_{answer} = c_{trial} = 0.15$ at ordinary costs; trial costs are additional and per party. Each incurs 0.30 through trial before fee shifting. No separate bargaining cost.
Cost multipliers	A common multiplier $c \in \{0.25, 0.5, 1, 2, 4\}$ scales all stage costs; ordinary costs use $c = 1$.
Preferences	Both parties maximize expected utility: risk neutral, $U(W) = W$, or CARA, $U(W) = -e^{-\alpha W}$ with $\alpha = 2$ for both parties.

		Trial	Complete
Fee trigger	American	Fee-Shifting	Fee-Shifting
After trial	Own costs	Loser pays	Loser pays
Initial nonanswer	Own costs	Own costs	D pays P's fees
Later P abandonment	Own costs	Own costs	P pays D's fees
Later D default	Own costs	Own costs	D pays P's fees
Settlement	Own costs	Own costs	Own costs

Figure 2 illustrates a portion of the game tree for the baseline simulation under the American rule. The C squares represent chance decisions, while the P and D circles represent plaintiff and defendant information sets numbered in the full game tree. Decision probabilities are rounded to three decimal places. For example, at D_{12} , the defendant does not know the plaintiff's signal and acts based on its anticipated utility absent this information, deciding to answer because that produces higher utility. Meanwhile, P_{48} illustrates a mixed strategy, where the plaintiff is indifferent whether to file because each decision yields the same wealth and thus utility. The second panel in the diagram continues the first and illustrates two successful settlements, which both happen to occur when the plaintiff's demand happens to equal the defendant's offer. Finally, trial probabilities are listed in the third panel.

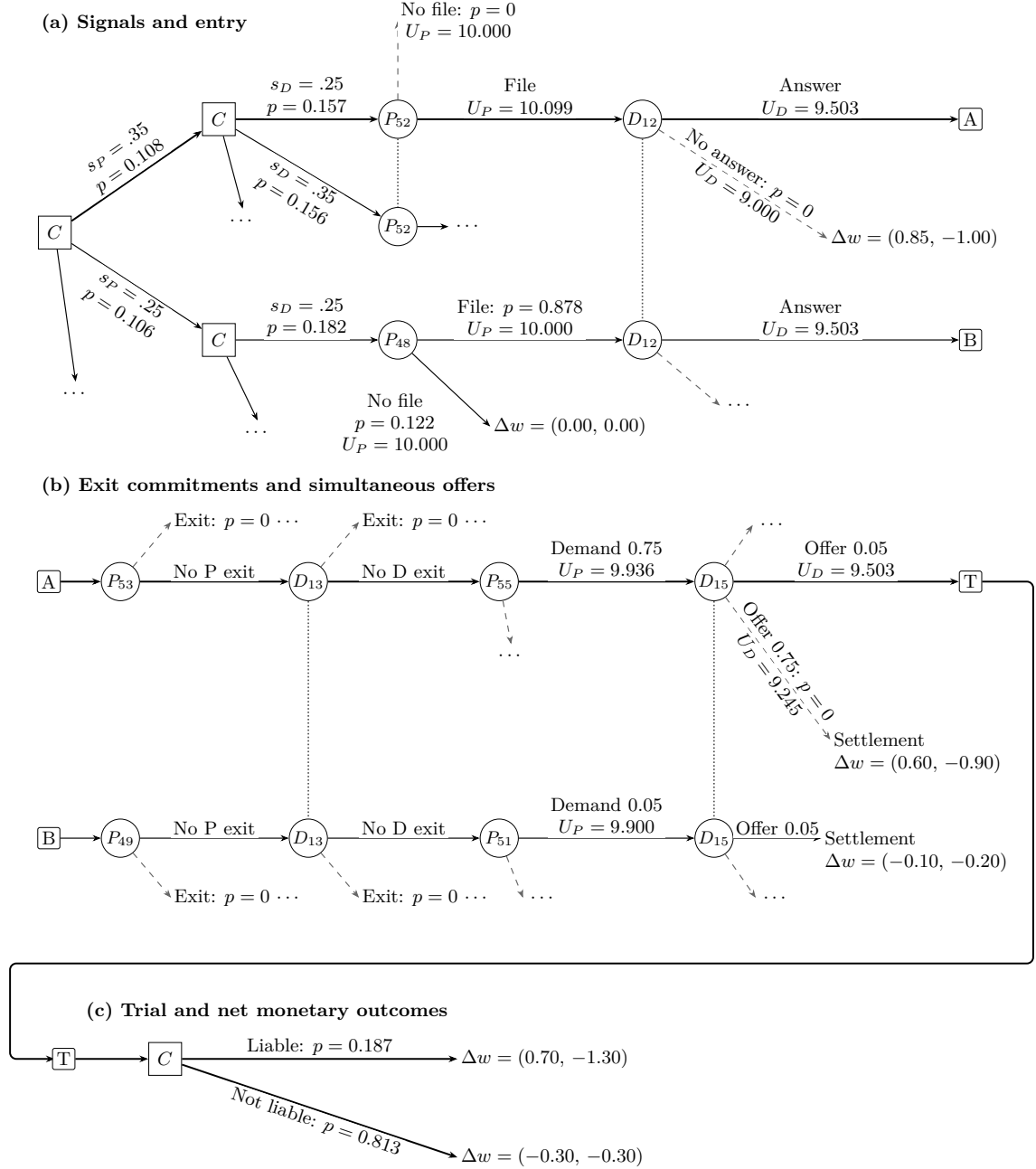


Figure 2: Worked equilibrium path

As a computational simplification with no effect on the equilibria, chance decisions at the end of games are collapsed. Replacing the court's decision with a single node reflecting the parties' expected utilities, taking into account risk aversion if applicable, leaves their incentives unchanged. Similarly, when both parties decide to quit, the expected utility from the mutual-withdrawal lottery can be calculated directly.

2.5 Outcome measures

Once an equilibrium is calculated, outcome probabilities and welfare measures can be evaluated by weighting the possible dispositions and underlying truth states. These are deterministic expectations rather than Monte Carlo estimates.

True liability T is binary, with $\Pr(T = 1 \mid Q = q) = q$.¹ Thus, although true liability and the strength of the plaintiff's case are correlated, a plaintiff can have a weak case despite true liability, or a strong case despite the absence of true liability. Neither party directly observes Q or T .

Explicit modeling of true liability permits measurement of the accuracy of outcomes. We can compare each party's net gain or loss with the outcome in a hypothetical perfect and costless litigation system. Defining accuracy solely by whether settlements mimic judgments would instead ignore imperfections in judgments themselves. A more ambitious approach would model the emergence of disputes endogenously in a particular legal context, as in Hylton (2002) [14], where a potential tort defendant initially decides how much care to take. This is left for a future extension, and the present model treats the population of potential disputes as given.

The monetary measures distinguish three interests. For a meritorious plaintiff, the shortfall is the amount by which net recovery, including litigation costs and fee transfers, falls below the unit damages award (or zero, if a full recovery is obtained). For a truly nonliable defendant, the burden is its net loss from litigation. For a truly liable defendant, the excess burden is its net loss above the unit damages award (or zero, if there is no net burden). Each measure is weighted by the probability of the relevant truth state, so that the reported figures are contributions per potential dispute, including disputes never filed.

Gross outcome error separately measures the absolute difference between the base payment and the payment warranted by true liability, before litigation costs and separately awarded fees. Real expenditures measure resources consumed in filing, answering, and trial. These measures distinguish errors in payments from the costs of obtaining a disposition. Their relative importance may be context-sensitive.

¹This can be derived by assuming that Chance first determines true liability, with $\Pr(T = 1) = \frac{1}{2}$, and then draws case quality from $Q \mid T = 1 \sim \text{Beta}(2, 1)$ or $Q \mid T = 0 \sim \text{Beta}(1, 2)$. These are probability distributions that have linear densities $2q$ and $2(1 - q)$, respectively. Their equal mixture is uniform on $[0, 1]$, and Bayes' rule yields $\Pr(T = 1 \mid Q = q) = q$. The Online Repository features extensions with different distributions of case merits.

3 Results

This part reports the principal results, focusing on the six baseline cases, involving three fee-shifting rules (American, trial fee shifting, and complete shifting) with both risk neutrality and risk aversion. Section 3.1 begins with the results under risk neutrality. In addition to showing the dispositions and the party strategies that produce them, it identifies the key explanations for the differences by examining best responses to opponents' equilibrium strategies under both rules. Section 3.2 similarly reports results for risk aversion. In both cases, trial-only fee shifting has a scant effect on filing and answering, but complete risk aversion increases rates at which plaintiffs fail to file while leading defendants to always answer. Section 3.3 considers the welfare implications of the two fee-shifting rules.

3.1 Risk Neutrality

Figure 3 reports the dispositions of potential disputes under the three fee-shifting rules with risk neutrality and ordinary costs. The American rule results show little settlement. In 19.1% of potential cases, the plaintiff does not file, and in 38.3%, the defendant defaults after the plaintiff files. A settlement occurs only 8.3% of the time, and the remaining 45.3% of potential cases go to trial, with the plaintiff winning slightly more than the defendant. It is true that one might consider some of the party defaults to be a form of settlement, particularly given that the model does not include a pre-filing settlement phase. Nonetheless, it may be surprising that conditional on filing and answering, relatively few cases go to trial, particularly given empirical data on filing rates. [CITE KLERMAN AND COAUTHOR] After all, the information structure was designed to give parties relatively good information about the strength of the plaintiff's claim, and parties that settle can split the surplus from such settlement.

The fee-shifting results may seem even more puzzling. First, switching from the American rule to trial fee-shifting considerably increases the rate of trial conditional on filing and answering, from 19.0% to 41.7%. This reverses the familiar finding that fee shifting should make trial more likely in litigated cases, because higher stakes should make mutually optimistic parties less willing to settle. [CITE] Second, switching to complete fee-shifting produces yet another increase, to 57.8% of cases conditional on filing and answering. And third, and perhaps most oddly, the proportion of cases in which potential plaintiffs do not file increases greatly (from 19.1% and 17.8% in the American and trial fee-shifting cases, respectively, to 39.2%), while the proportion of all cases in which the defendant does not file tumbles (from 37.3% and 38.3%, respectively, to 0).

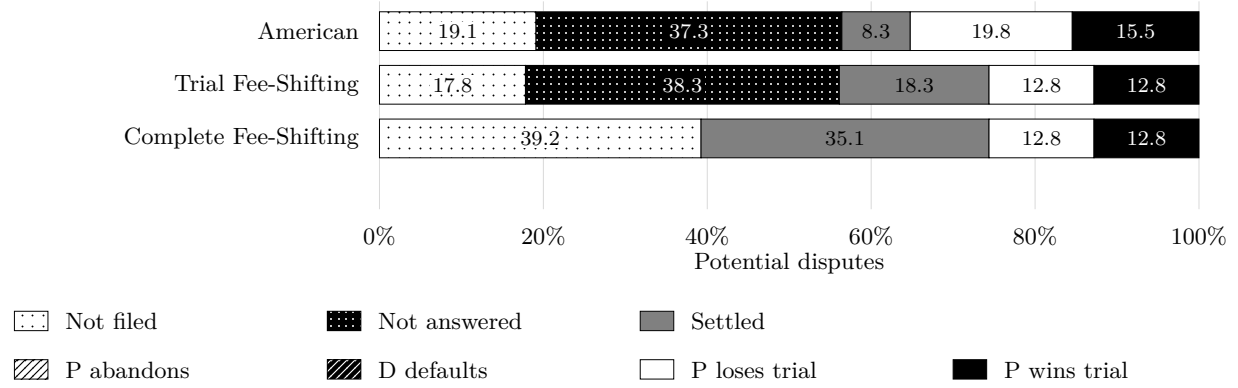


Figure 3: Risk-neutral dispositions

That strategic bargaining might occur is not surprising. All of the Bayesian models of settlement that use Chatterjee-Samuelson bargaining (including Friedman and Wittman (2006) [9]) recognize that each party faces a trade-off: Greater generosity increases the chance of avoiding trial, but greater stinginess increases the portion of the bargaining surplus that the offeror will receive if a settlement does occur. But strategic bargaining plays a considerably larger role than one might expect. Figure 4 illustrates substantial strategic bargaining even with the baseline information structure. Real-world bargaining includes many features not accounted for in this model, but this result tentatively suggests that litigants might rationally often fail to reach settlements even when their actual assessments of the litigation value are not far apart. At least, these results should make us cautious about accepting at face value the results of Landes-Posner-Gould models that assume settlement occurs whenever settlement would produce social surplus.

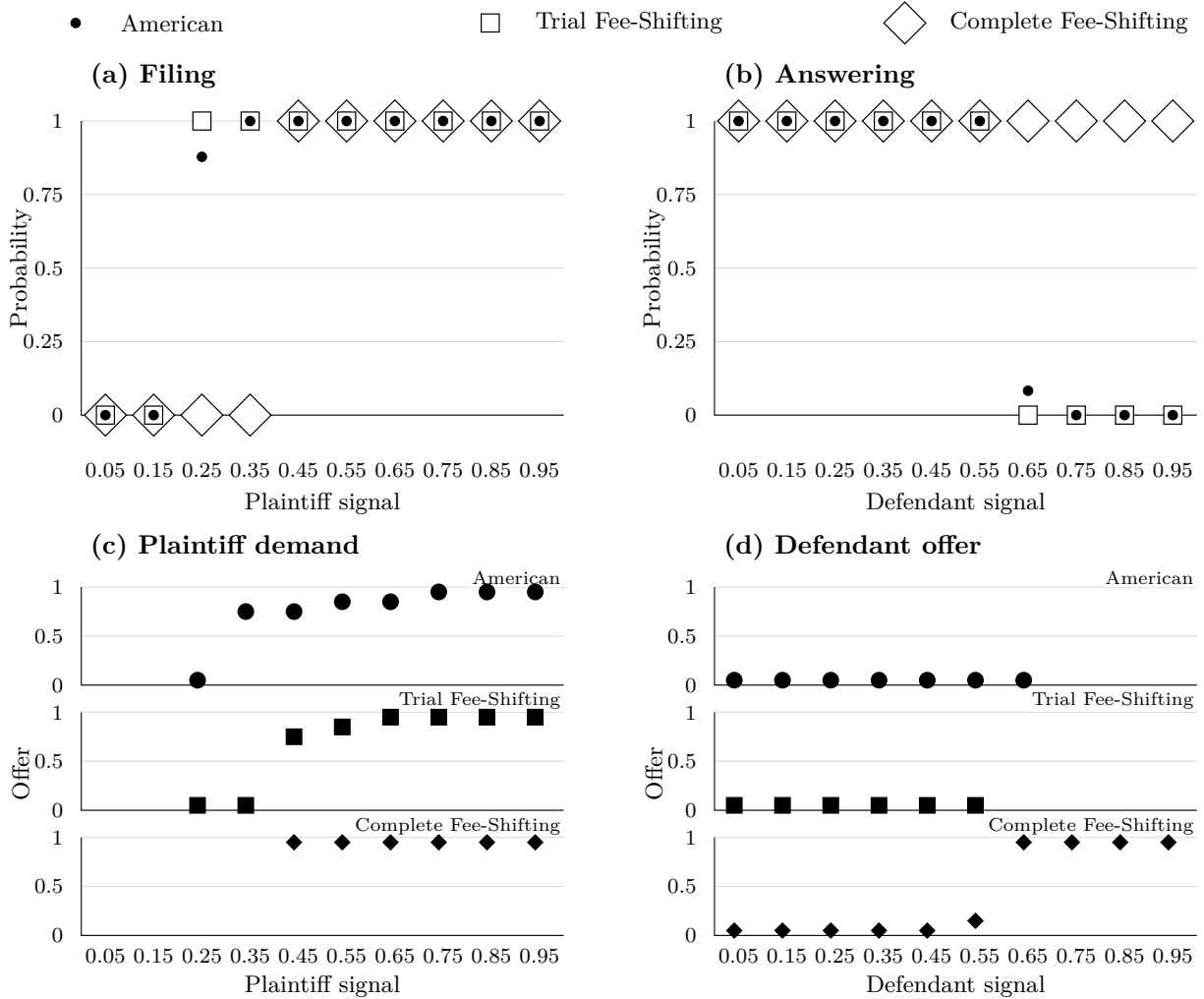


Figure 4: Risk-neutral participation and offers

For many signals, the parties' strategies are the same under all rules. For example, the plaintiff never files with a signal of 0.05 or 0.15 and always files with a signal of 0.45 or higher. Similarly, the defendant always answers with a signal up to 0.55. But at signals 0.25 and 0.35, the plaintiff does not file with complete fee shifting, while filing a high percentage or all of the time with the other two rules, and the same occurs on the defendants' side. The increase in settlements, meanwhile, arises principally from the signals of 0.65 or higher in which the defendant answers only in the complete fee-shifting regime. In those cases, the defendant offers settlements of 0.95. That leads to completed settlements at that same value when the plaintiff offers settlements of 0.45 or higher. The plaintiff insists on a settlement at that value with a signal of 0.45 or greater, whereas, with the American rule and trial fee-shifting, the plaintiff's offer was that high only for a signal of 0.75 or 0.65, respectively.

The one-sided nature of settlements extends across all three rules. With the American rule, the only

settlements occur when the plaintiff's signal is 0.25, and the plaintiff accepts what might be seen as a nuisance payoff of 0.05. Under trial fee-shifting, the settlements occur at that signal and 0.35 for the same value. Complete fee-shifting simply changes the value at which settlements occur to the other side of the probability spectrum. More generally, across all rules, each party insists on a settlement highly favorable to it under most signals, conditional on filing and entering, but for relatively extreme values will accede to a settlement highly favorable to the other party. It turns out to be a better strategy for a litigant to hold out for the highly favorable settlements than to offer middling settlements that would more evenly split the apparent surplus from settlement.

To better appreciate the logic underlying changes across equilibria, we need to assess whether changes in strategies occur as a direct result of a change in the fee-shifting rule or as a result of an opponent's change in strategy, itself as a result of a change in the fee-shifting rule, or as some combination. The exercise proves straightforward for the three strategies at baseline. Table 2 presents selected decompositions of the changes in equilibrium strategies under the fee-shifting rules. When changing from the American rule to trial fee-shifting, the most important change occurs in the plaintiff demand function at signal 0.35, with the plaintiff dropping the demand from 0.75 to the nuisance value of 0.05. This drop occurs even if we calculate the plaintiff's optimal strategy while holding the defendant's strategy constant at its strategy under the American rule. Thus, we can say with confidence that the entire drop is a direct result of the change in rules.

With the change from trial to complete fee-shifting, however, the story is more complex. The defendant's decision to answer across a range of high signals can be entirely directly attributable to the change in rules, but none of the plaintiff's decision not to file at signals 0.25 and 0.35 can be attributed to the change in rules. Rather, the change is entirely attributable to the defendant's decision to answer. When the defendant's strategy is held constant at the strategy under trial fee shifting but applying the complete fee-shifting rule, the plaintiff does not change its file decision, but if the defendant's strategy is altered to include the defendant's offers under the complete fee-shifting equilibrium, then the plaintiff does change its file decision. Once we know that the change in equilibrium is set off entirely by the defendant's decision to always answer, the underlying motivating logic is simple. A defendant who does not answer must pay the plaintiff's filing fee, while the defendant can potentially avoid that expense by continuing to litigate.

Table 2: Selected risk-neutral strategy changes

Decision	Signal	Original $\rightarrow$ Target	Direct	Opponent entry	Opponent offers
<i>A. American to Trial Fee-Shifting; risk neutral</i>					
P demand, continue	0.35	0.75 $\rightarrow$ 0.05	-0.70	0.00	0.00
<i>B. Trial to Complete Fee-Shifting; risk neutral</i>					
D answers	0.65-0.95	0% $\rightarrow$ 100%	+100	0	0
P files	0.25-0.35	100% $\rightarrow$ 0%	0	-100	0

Note: Direct changes the fee rule while holding the opponent's strategy fixed. Opponent-entry and opponent-offer contributions average over the orders of replacing opponent strategy components. Filing and answering contributions are percentage points; demand contributions are fractions of damages. Opponent-exit and remaining contributions are zero in these rows. The full analysis and diagnostic checks are in the online repository.

3.2 Risk Aversion

Figure 5 reports dispositions when both parties have CARA risk aversion with coefficient $\alpha = 2$, holding ordinary costs and the information structure fixed. Figure 6 shows the corresponding participation and offer strategies. Table 3 presents selected strategy changes from introducing risk aversion and from extending fee shifting under risk aversion.

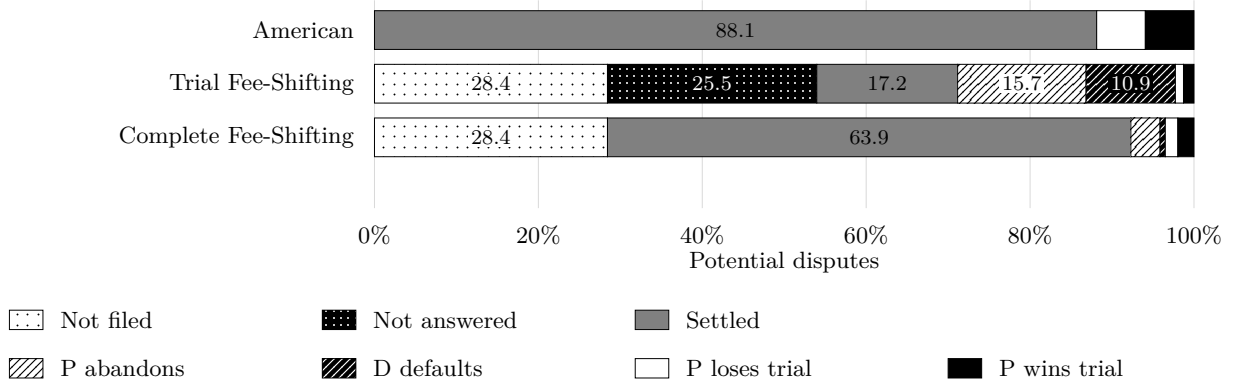


Figure 5: Risk-averse dispositions

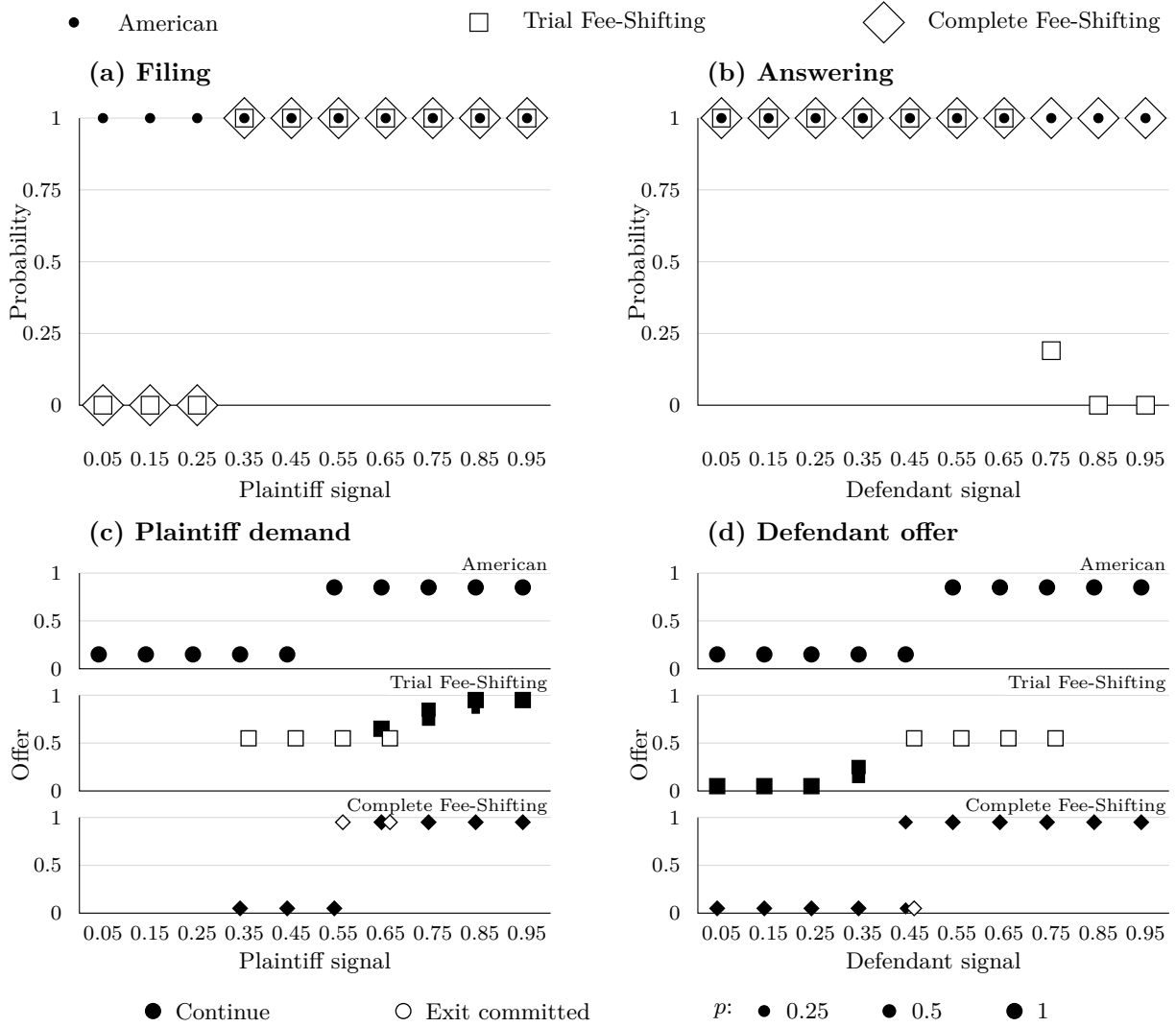


Figure 6: Risk-averse participation and offers

Table 3: Selected strategy changes involving risk aversion

Decision	Signal	Original → Target	Direct	Opponent entry	Opponent offers	Opponent exit
<i>A. Risk neutral to risk averse; American</i>						
P files	0.05–0.15	0% → 100%	0	0	+100	0
P files	0.35	100% → 100%	−100	0	+100	0
P demand, continue	0.35	0.75 → 0.15	−0.70	0.00	+0.10	0.00
D offer, continue	0.05–0.15	0.05 → 0.15	0.00	0.00	+0.10	0.00
D offer, continue	0.65	0.05 → 0.85	+0.90	0.00	−0.10	0.00
<i>B. Risk neutral to risk averse; Trial Fee-Shifting</i>						
P commits to exit	0.45	0% → 100%	0	0	0	+100
D commits to default	0.45	0% → 100%	+100	+16.7	+16.7	−33.3
D commits to default	0.55	0% → 100%	0	0	+50	+50
<i>C. Risk neutral to risk averse; Complete Fee-Shifting</i>						
P demand, continue	0.45–0.55	0.95 → 0.05	−0.90	0.00	0.00	0.00
<i>D. Trial to Complete Fee-Shifting; risk averse</i>						
D answers	0.75	19% → 100%	+81	0	0	0
D answers	0.85–0.95	0% → 100%	+100	0	0	0

Note: Panels A–C change both parties’ risk preferences; panel D changes the fee rule. Contributions are defined as in Table 2, with opponent-exit contributions also shown. Demand and offer amounts and contributions are fractions of damages. Displayed histories are reached in both equilibria. The exit commitment applies after bargaining failure; demands and offers condition on commitment to continue. Remaining contributions are zero. Full results and diagnostic checks are in the online repository.

Consider first the effect of adding risk aversion, holding constant the fee-shifting rule. With the American rule, settlement becomes the dominant method of resolution, occurring in 88.1% of cases. Interestingly, however, the players’ strategies still do not embody gradual increases in offers based on signals. Instead, each party’s strategy pools at the 0.15 level for signals up to 0.45 and 0.85 for higher signals. Table 3 highlights that some of these offer changes (such as the defendant offering 0.85 instead of 0.05 at signal level 0.65 and the plaintiff demanding 0.15 instead of 0.75 at the mirror signal 0.35) are predominantly direct consequences of risk aversion. The risk of trial leads parties to become generous at signals that indicate slight weakness in their case. The more interesting change is in filing and answering. The plaintiff now always files for signals less than or equal to 0.35, and what motivates this change is not the direct effect of risk aversion, but the change in defendant offers. With signal 0.35, the direct effect of risk aversion is that the plaintiff switches from risky and uncertain filing to not filing, but the increased generosity of the defendant’s offers entirely counters this.

When risk aversion is added to the trial fee-shifting baseline, the possibility of late exit emerges as an

important factor. Sometimes, it is worthwhile for a party to file or answer and hope for an advantageous settlement, but if settlement fails, then exit is more attractive than trial. This is especially so when fees incurred during trial are shifted. Table 3 illustrates that introducing risk aversion under trial fee shifting directly induces the defendant at signal 0.45 to commit to default if settlement fails. Because each player has offer strategies contingent on the player's exit decision, this changes the defendant's offer strategies in turn, and it also becomes more profitable for the plaintiff to exit. Finally, adding risk aversion to complete fee shifting leads to a smaller incidence of exit. Instead, the most notable consequence is that the plaintiff offer when it continues instead of exiting falls from 0.95 to just 0.05, part of a pattern of greatly increased settlement.

Yet the most revealing decomposition in Table 3 is in panel D, comparing trial to complete fee-shifting, both under risk aversion. The defendant now answers all the time, even when its signal corresponds to a strong plaintiff case. This change is mostly a direct consequence of the change in the fee-shifting rule. The logic is as above, that with complete feeshifting, once the plaintiff enters, answering provides is the only means that the defendant has to potentially escape paying the fees the plaintiff has already incurred for filing. Sinking fees that the other party may need to pay can provide that party a strong incentive to keep litigating. With risk aversion, however, this continued fighting is more likely to result in a settlement or exit than in a trial.

3.3 Welfare Analysis

Table 4 reports welfare measures under both risk neutrality and risk aversion. It compares monetary burdens, gross outcome error, and real expenditures across fee rules and risk preferences, using the definitions in Section 2.5. Because complete fee-shifting promotes defendant non-answer with both risk neutrality and risk aversion, the effects of fee-shifting on the average plaintiff and the defendant are not identical. The first two columns of Table 4 contain measures of the quality of the litigation system from the perspective of the party in the right, a meritorious plaintiff in the first column, and a nonliable defendant in the second. These measures may be more useful than the gross outcome accuracy and total expenditures figures in the fourth and fifth columns, because they combine considerations of accuracy and litigation expense to determine how they in combination may adversely affect innocent parties.

When the meritorious plaintiff falls short of full recovery, taking into account its litigation expenses, or the nonliable defendant bears a burden from either litigation expenses or the judgment, the legal system's imperfections impose costs on innocent parties. Under both risk neutrality and risk aversion, the average meritorious plaintiff prefers trial fee-shifting to complete fee-shifting, given the tendency of the latter to

lead to defendant non-answer. For the nonliable defendant, the preference is reversed under risk neutrality, thought the relative burdens are nearly identical under risk aversion. Meanwhile, a liable defendant bears a greater excess burden (total spending above the level of damages) with complete fee-shifting under both risk preferences. Whether this represents an improvement in deterrence or overdeterrence may depend on the particular litigation context.

Table 4: Welfare outcomes under risk neutrality and risk aversion

Risk neutral					
Fee rule	Meritorious-P shortfall	Nonliable-D burden	Liable-D excess burden	Gross outcome error	Total expenditures
American	0.216	0.222	0.028	0.278	0.292
Trial Fee-Shifting	0.209	0.198	0.047	0.277	0.266
Complete Fee-Shifting	0.232	0.165	0.075	0.285	0.259
Risk averse					
Fee rule	Meritorious-P shortfall	Nonliable-D burden	Liable-D excess burden	Gross outcome error	Total expenditures
American	0.256	0.256	0.011	0.344	0.336
Trial Fee-Shifting	0.224	0.164	0.017	0.286	0.183
Complete Fee-Shifting	0.241	0.166	0.038	0.298	0.225

4 Robustness and Limits

4.1 Cost Levels and Timing

The parameter values are not calibrated to actual data, and real-world values are likely to vary from place to place and case category to case category. An important dimension is the cost relative to the stakes. Litigation may play out very differently in a small claim, in which lawyers' fees and the parties' own time may be large relative to the amount at issue, than in a large suit in which legal expenses constitute a small fraction of the stakes. To examine this dimension, the baseline cost levels are multiplied by each element of $\{\frac{1}{4}, \frac{1}{2}, 1, 2, 4\}$, holding the other features of each specification fixed.

Settlement bargaining dynamics depend on when costs are incurred. The cost-timing extensions hold constant the total costs that each party would bear if a case reached trial, but vary whether those costs are borne at entry or remain avoidable until trial. These comparisons change the distribution of fixed costs across stages; they do not model the parties' choice of how much to spend.

4.2 Information Quality and Structure

The information-quality checks vary the parties' and court's signal noise together and consider alternative distributions of case quality and relationships between true liability and signals. The full comparisons are reported in the online repository.

4.3 Offer-Grid Sensitivity

The granularity of available offers is another potential source of sensitivity. Suppose, for example, that a change in some parameter would lead in a continuous model to slight changes in the parties' settlement functions that nonetheless had large effects on the settlement rate. It could be difficult for a discrete model, in which each party must choose among only ten offer levels, to capture that nuance. Discrete offers are therefore a limitation even when the selected equilibrium features widely separated demands and offers. Comparing ten and fifteen available offers tests sensitivity to a finer grid, but does not establish convergence to a model with continuous offers.

4.4 Multiple Equilibria

Finding an equilibrium does not establish whether it is representative of other equilibria of the same game. The computational approach therefore raises the question whether the results depend on which equilibrium the algorithm finds. I address this by varying the initial strategy profiles supplied to the solver and examining both the distinct profiles recovered and the variation in their outcomes. This exercise cannot determine how many equilibria exist: in principle, the recovered profiles could be the entire population or a small sample. Recovery frequencies describe the numerical search rather than the probability that litigants would select a particular equilibrium.

5 Conclusion

At least three limitations affect the interpretation of the model. First, litigants may not be rational Bayesians who play equilibrium strategies. But, as the literature generally presumes, models assuming rationality may provide a first guess at how litigants behave, and economic forces may gradually push actors in any given legal context toward equilibrium strategies. Moreover, though more complicated to model, two-sided asymmetric information seems like a more accurate description of most litigation than one-sided information, and a model that recognizes the possibility that litigants may not contest litigation or may quit after settlement failure illustrates dynamics that some models assume away.

Second, models' usefulness may depend on parameter values, and the parameters here are not calibrated based on data, either statistics from actual litigation or on microfoundations of individual behavior. Different categories of litigation may require different calibrations. As Gelbach (2018) [10] observes, quite different structural models may also share a common reduced form, making empirical discrimination difficult.

Third, the model omits important features of litigation, including learning and bargaining over time, as well as the role of lawyers and agency costs. Indeed, this is but a first step toward building models that can incorporate even more complex features of the litigation game. Because the game tree size may increase exponentially with additional features, other computational game theory approaches that seek approximate equilibria may be needed to make larger game models solvable.

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- A Model and computational details
- B Alternative information structures
- C Costs, extensions, and strategy mechanisms
- D Equilibrium and discretization checks
- E Data and reproducibility